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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2021.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — ( $10 \times 2 = 20$  marks)

Answer ALL the questions.

All questions carry equal marks.

1. Find the partial differential equation of all planes cutting equal intercepts from the  $x$  and  $y$  axis.
2. Find the general solution of  
 $p \tan x + q \tan y = \tan z$ .
3. Write down the partial differential equation governing the transverse vibrations of an elastic string.
4. What conditions are assumed in deriving the one-dimensional wave equation?

5. Write all variable separable solutions of the one dimensional heat equation.
6. Derive the solution of one dimensional heat flow equation under the steady state conditions.
7. State the principle of least squares.
8. Define type I and type II error.
9. Mention any two properties of t -distribution.
10. State the conditions for the validity of  $\chi^2$  - test.

SECTION B — (5 × 11 = 55 marks)

Answer ONE full question from each Unit.

#### UNIT I

11. Solve  $z = px + qy + \sqrt{1 + p^2 + q^2}$ .

Or

12. Solve  $(3z - 4y)p + (4x - 2z)q = 2y - 3x$ .

#### UNIT II

13. Solve  $2x \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial y} = 0$  by method of separation of variables.

Or

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14. A tightly stretched string with fixed end points  $x = 0$  and  $x = l$  is initially in a position given by  $y(x, 0) = y_0 \sin^3\left(\frac{nx}{l}\right)$ . If it is released from the rest of this position, find the displacement  $y$  at any point and at any distance from the end  $x = 0$ .

#### UNIT III

15. A bar of length 10cm, long, has its ends A and B kept at  $50^\circ\text{C}$  and  $100^\circ\text{C}$ , until steady-state conditions prevail. The temperature at A is then suddenly raised to  $90^\circ\text{C}$  and at the same instant, that at B is lowered to  $60^\circ\text{C}$  and the end temperature are maintained thereafter. Find the temperature at distance  $x$  from end A at time  $t$ .

Or

16. A rectangular plate is bounded by the lines  $x = 0$ ,  $y = 0$ ,  $x = a$  and  $y = b$ . Its surfaces are insulated and the temperature along two adjacent edges are kept at  $100^\circ\text{C}$ , while the temperature along the other two edges are at  $0^\circ\text{C}$ . Find the steady-state temperature at any point in the plate.

#### UNIT IV

17. The data given below will fit a formula of the type  $y = a + bx + cx^2$ . Find the formula.

$x$ : 87.5   84.0   77.8   63.7   46.7   36.9

$y$ : 292   283   270   235   197   181

Or

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18. Before an increase in exercise duty on tea, 800 people out of a sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether there is significant decrease in the consumption of tea after the increase in duty.

#### UNIT V

19. The heights of ten males of a given locality are found to be 175, 168, 155, 170, 152, 170, 160, 160 and 165 cms. Based on the sample, find the 95% confidence limits for the height of males in that locality.

Or

20. Theory predicts that the proportion of beans in four groups A, B, C, D should be 9:3:3:1. In an experiment among 1600 beans, the numbers in the four groups were 882, 313, 287 and 118. Does the experiment support the theory?



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B.Tech. DEGREE EXAMINATION, JANUARY 2023.

Fourth Semester

MATHEMATICS – IV

(From 2013-14 onwards)

(Common to All Branches)

Time : Three hours

Maximum : 75 marks

PART A — ( $10 \times 2 = 20$  marks)

Answer ALL questions.

1. Eliminate  $a$  and  $b$  from  $Z = ax + by + 9$ .
2. Define general integral
3. Write the one — dimensional wave equation.
4. Explain singular integral.
5. Explain the initial and boundary conditions.
6. Write down the equation of heat flow in two dimensions in polar coordinates.
7. Write the normal equations to fit a straight line  $y = a + bx$ .
8. Write the formula to test the difference of proportions in large samples.



9. Write any two applications of  $F$  — distribution.
10. Write down the formula for testing the difference of means and correlation coefficient.

PART B — ( $5 \times 11 = 55$  marks)

Answer ALL the units, Choosing ONE full question from each unit.

#### UNIT I

11. (a) Eliminate  $f$  and  $\phi$  from the relation  $z = f(x+ay) + \phi(x-ay)$ .
- (b) Find the integral surface of  $x^2p + y^2q + z^2 = 0$ , which passes through the hyperbola  $xy = x + y, z = 1$ .

Or

12. (a) Solve  $(D^2 + 16)y = e^{-3x} + \cos 4x$ .
- (b) Solve  $(D^2 + 3DD' + 2D'^2)z = \sin(x+5y)$ .

#### UNIT II

13. A string is stretched and fastened to two points  $l$  apart. Motion is started by displacing the string into the form  $y = y_0 \sin \frac{\pi x}{l}$  from which it is released at time  $t = 0$ . Find the displacement at time  $t$ .

Or

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14. A taut string of length  $2l$  is fastened at both ends. The midpoint of the string is taken to a height  $b$  and then released from rest in that position. Find the displacement of the string at time  $t$ .

#### UNIT III

15. Find the solutions of the wave equation that satisfies the following boundary and initial conditions  $y(0, t) = y(2\pi, t) = y_t(x, 0) = 0$ ;  $y(x, 0) = \sin 2\pi x$ .

Or

16. The vertices of a thin square plate are  $(0,0)$ ,  $(l, 0)$ ,  $(0,l)$ , and  $(l,l)$ . The upper edge of the square is maintained at an arbitrary temperature distribution given by  $u(x, l) = f(x)$ . The other three edges are maintained at the temperature zero. Find the steady-state temperature at any point within the plate.

#### UNIT IV

17. Fit a straight line  $y = Ax + B$  for the folk data.
- |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| X: | 0  | 5  | 10 | 15 | 20 | 25 | 30 |
| Y: | 10 | 14 | 19 | 25 | 31 | 36 | 39 |

Or

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18. (a) A sample of 19 members has a mean 3.4 cm., and standard deviation 2.61 cm.,. Is the sample from a large population of mean 3.25 cm., and standard deviation 2.61 cm?
- (b) The meaning of two single large samples of 1,000 and 2,000 members be regarded as drawn from the same population of standard deviation 2.5 inches. Test at 5% level of significance.

### UNIT V

19. (a) The following table gives the number air craft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

|                   |     |     |     |      |     |     |
|-------------------|-----|-----|-----|------|-----|-----|
| Days :            | Mon | Tue | Wed | Thus | Fri | Sat |
| No of Accidents : | 14  | 18  | 12  | 11   | 15  | 14  |

- (b) A coefficient of correlation of 0.2 is derived from a random sample of 625 pairs of observations. Is the significant?

Or

20. The heights of six randomly chosen sailors are in inches: 63, 62, 65, 66, 69, 69, 70, 71, 72. Those of 10 randomly chosen soldiers are 61, 62, 65, 66, 69, 69, 70, 71, 72 and 73. Test whether the sailors are on the average taller than soldiers.